

Homework 3

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Lemma A. $f \simeq g \implies \bar{f} \simeq \bar{g}$

Proof. $\forall g, f : [0, 1] \rightarrow X$ where $g \simeq f$. Thus $\exists F : I \times I \rightarrow Y \ni f_t(x) = F(x, t)$ and $f_0(x) := f(x)$ and $f_1(x) := g(x)$. Let $G : X \times I \rightarrow Y \ni G(x, t) = F(1 - x, t)$ and $g_t(x) := G(x, t)$. $\forall x \in I$, $g_0(x) = F(1 - x, 0) = f_0(1 - x) = f(1 - x) = \bar{f}(x)$, similarly, $g_1(x) = F(1 - x, 1) = f_1(1 - x) = g(1 - x) = \bar{g}(x)$. Thus there is a homotopy between $\bar{f}$ and $\bar{g}$. Thus $\bar{f} \simeq \bar{g}$. $\square$

Problem 1. Show the composition of paths satisfies the following cancellation property: If $f_0 \cdot g_0 \simeq f_0 \cdot g_1$ and $g_0 \simeq g_1$ then $f_0 \simeq f_1$

Solution. Suppose $f_0 \cdot g_0 \simeq f_0 \cdot g_1$ and $g_0 \simeq g_1$. Since $g_0 \simeq g_1$, that means $\bar{g}_0 \simeq \bar{g}_1$ by Lemma A.

$$f_0 \simeq f_0 \cdot c \simeq f_0 \cdot (g_0 \cdot \bar{g}_0) \simeq (f_0 \cdot g_0) \cdot \bar{g}_0 \simeq (f_1 \cdot g_1) \cdot g_0 \simeq f_1 \cdot g_1 \cdot \bar{g}_0 \simeq f_1 \cdot (g_1 \cdot \bar{g}_1) \simeq f_1 \cdot c \simeq f_1$$

Thus $f_0 \simeq f_1$ ■

Problem 2. Show that the change of base point homomorphism β_h depends only on the homotopy class of h

Solution. Let $\beta_{h_1}, \beta_{h_2} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ be change of base point homomorphisms given by paths h_1 and h_2 . Suppose $h_1 \simeq h_2$, then $\bar{h}_1 \simeq \bar{h}_2$. Then $\forall f \in \pi_1(X, x_1)$, $h_1 \cdot f \cdot \bar{h}_1 \simeq h_2 \cdot f \cdot \bar{h}_2$. Thus $\beta_{h_1}(f) = h_1 \cdot f \cdot \bar{h}_1 \simeq h_2 \cdot f \cdot \bar{h}_2 = \beta_{h_2}(f)$. Thus if two paths are homotopic then the homomorphisms they induce are the same and differ only based on choice of base point. Thus the change of base point homomorphism β_h depends only on the homotopy class of h . ■

Problem 3. For a path-connected space X , show that $\pi_1(X)$ is abelian iff all basepoint-change homomorphisms β_h depend only on the endpoints of the path h .

Proof. $\implies$: Suppose $\pi_1(X)$ is abelian. Suppose $\beta_{h_1}, \beta_{h_2} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ are basepoint change homomorphisms with the same base points, x_0, x_1 where h_1 and h_2 are paths from x_0 to x_1 . Note the following thing, that $(h_2 \cdot \bar{h}_1)(0) = h_2(0) = x_0$, $x_1 = \bar{h}_1(0) = (h_2 \cdot \bar{h}_1)(\frac{1}{2}) = h_2(1) = x_1$, and $(h_2 \cdot \bar{h}_1)(1) = \bar{h}_1(1) = x_0$. Thus $h_2 \cdot \bar{h}_1$ is a loop about x_0 . Thus, $\forall f \in \pi_1(X, x_1)$

$$\begin{aligned}
\beta_{h_1}(f) &= h_1 \cdot f \cdot \overline{h_1} \\
&= h_1 \cdot f \cdot c \cdot \overline{h_1} \\
&= h_1 \cdot f \cdot (\overline{h_2} \cdot h_2) \cdot \overline{h_1} \\
&= (h_1 \cdot f \cdot \overline{h_2}) \cdot (h_2 \cdot \overline{h_1}) \\
&= (h_2 \cdot \overline{h_1}) \cdot (h_1 \cdot f \cdot \overline{h_2}) \quad (\text{Since } \pi_1(X) \text{ is abelian thus } \pi_1(X, x_0) \text{ is abelian}) \\
&= h_2 \cdot (\overline{h_1} \cdot h_1) \cdot f \cdot \overline{h_2} \\
&= h_2 \cdot f \cdot \overline{h_2} = \beta_{h_2}(f)
\end{aligned}$$

$\Leftarrow$: Suppose all basepoint-change homomorphisms β_h depend only on the endpoints of the path h . $\forall x_0 \in X, \forall f, g \in \pi_1(X, x_0)$. Since g and c are paths which share the same endpoints namely x_0 on either end. $g \cdot f \cdot \overline{g} = \beta_g(f) = \beta_c(f) = c \cdot f \cdot \overline{c} = f$. Thus $g \cdot f \cdot \overline{g} = f \implies g \cdot g = f \cdot g$. Thus $\pi_1(X, x_0)$ is abelian. Since X is path connected, $\forall x_1$ there is a homotopy class of paths, h , connecting x_1 and x_0 , thus $\exists \beta_h$ a change of basis homomorphism, which is an isomorphism by Proposition 1.5. Thus since $\pi_1(X, x_0)$ is abelian, $\pi_1(X, x_1)$ is abelian. Thus $\pi_1(X, x_0) \cong \pi_1(X)$ is abelian. $\square$

Lemma B. *If X is a star shaped neighborhood with x_0 being its corresponding starshaped point, and $\exists f : I \rightarrow X$ a path contained in X . Let L^0 be the line from $f(0)$ to x_0 and L^1 be the line from $f(1)$ to x_0 . $f \simeq L^0 \cdot \overline{L^1}$.*

Proof. Let $L : I^2 \rightarrow X$ where $(s, t) \rightarrow (1-t)f(s) + tx_0$ and let $L_t := I \rightarrow X$ where $s \rightarrow L(s, t)$ and let $L^s := I \rightarrow X$ where $t \rightarrow L(s, t)$. Let $\alpha : I^2 \rightarrow X$ where $(s, t) \rightarrow L_{st}(0)$ where $\alpha_t : I \rightarrow X$ is $s \rightarrow \alpha(s, t)$. Let $\beta : I^2 \rightarrow X$ where $(s, t) \rightarrow L_{st}(1)$ where $\beta_t : I \rightarrow X$ is $s \rightarrow \beta(s, t)$. Let $F : I^2 \rightarrow X$ be the following map $F(s, t) = (\alpha_t \cdot L_t \cdot \overline{\beta_t})(s)$ or equivalently:

$$F(s, t) := \begin{cases} (1-2st)f(0) + 2tsx_0 & \text{if } s \in [0, \frac{1}{2}] \\ (1-t)f(4s-2) + tx_0 & \text{if } s \in [\frac{1}{2}, \frac{3}{4}] \\ (1-(4-4s)t)f(1) + (4-4s)tx_0 & \text{if } s \in [\frac{3}{4}, 1] \end{cases}$$

Thus $\forall s \in I$,

$$F(s, 0) := \begin{cases} f(0) & \text{if } s \in [0, \frac{1}{2}] \\ f(4s-2) & \text{if } s \in [\frac{1}{2}, \frac{3}{4}] \\ f(1) & \text{if } s \in [\frac{3}{4}, 1] \end{cases}$$

which means $F(s, 0) = (c_{f(0)} \cdot f \cdot c_{f(1)})(s)$. Similarly,

$$F(s, 1) := \begin{cases} (1-2s)f(0) + 2sx_0 & \text{if } s \in [0, \frac{1}{2}] \\ x_0 & \text{if } s \in [\frac{1}{2}, \frac{3}{4}] \\ (4s-3)f(1) + (4-4s)x_0 & \text{if } s \in [\frac{3}{4}, 1] \end{cases}$$

thus $F(s, 1) = (L^0 \cdot c_{x_0} \cdot \overline{L^1})(s)$.

Finally, $F(0, s) = (1-2s(0))f(0) + 2s(0)x_0 = f(0)$ and $F(1, s) = (1-(4-4s)s)f(1) + (0)sx_0 = f(1)$. Thus F is a homotopy which preserves end points.

Thus $c_{f(0)} \cdot f \cdot c_{f(1)} \simeq L^0 \cdot c_{x_0} \cdot \overline{L^1}$. But $c_{f(0)} \cdot f \cdot c_{f(1)} \simeq f$ and $L^0 \cdot c_{x_0} \cdot \overline{L^1} \simeq L^0 \cdot \overline{L^1}$. Thus $f \simeq L^0 \cdot \overline{L^1}$ $\square$

Problem 4a. A subspace $X \subseteq \mathbb{R}^n$ is said to be star-shaped if there is a point $x_0 \in X$ such that, for each $x \in X$, the line segment from x_0 to x lies in X . Show that if a subspace $X \subseteq \mathbb{R}^n$ is locally star-shaped, in the sense that every point of X has a star-shaped neighborhood in X , then every path in X is homotopic in X to a piecewise linear path, that is, a path consisting of a finite number of straight line segments traversed at constant speed.

Solution. Let $f : [0, 1] \rightarrow X$ be a path in X . $\forall x \in X$, let U_x be the star-shaped neighborhood around x , it exists since X is locally star-shaped. $\{U_x\}_{x \in X}$ is an open cover of X , thus $\{f^{-1}(U_x)\}_{x \in X}$ is an open cover of I . By the compactness and connectedness of I , $\exists$ a finite subcover $f^{-1}(U_{x_0}), \dots, f^{-1}(U_{x_n})$ of I such that $0 \in f^{-1}(U_{x_0})$ and $1 \in f^{-1}(U_{x_n})$ and $f^{-1}(U_{x_i}) \cap f^{-1}(U_{x_{i+1}}) \neq \emptyset$ for $i < n$. Note $\{x_i \mid 0 \leq i \leq n\}$ is the set of star-shaped points. Let $t_0, \dots, t_{n+1} \in I$ such that for $n \in [1, \dots, n]$, $t_0 = 0$, $t_i \in f^{-1}(U_{x_i}) \cap f^{-1}(U_{x_{i-1}})$, $[t_{i-1}, t_i] \subseteq f^{-1}(U_{x_{i-1}})$, $t_{n+1} = 1$, $[t_n, t_{n+1}] \subseteq f^{-1}(U_{x_n})$.

Let $\forall i \in [0, \dots, n]$, $f_i : I \rightarrow X$ where $s \rightarrow f|_{[t_i, t_{i+1}]}((t_{i+1} - t_i)s + t_i) = f((t_{i+1} - t_i)s + t_i)$. Note $f \simeq f_0 \dots f_n$. Furthermore since $f_i \in U_{x_i}$, by Lemma B $f_i \simeq L_{f(t_i) \rightarrow x_i} \cdot \overline{L_{f(t_{i+1}) \rightarrow x_i}}$, where $L_{p \rightarrow q}$ is the straight line function from point p to q . Since U_{x_i} is the star-shaped neighborhood around x_i , $L_{f(t_i) \rightarrow x_i}(I)$ and $L_{f(t_{i+1}) \rightarrow x_i}(I)$ are completely contained within U_{x_i} .

$$\begin{aligned} f &\simeq f_0 \dots f_n \simeq L_{f(t_0) \rightarrow x_0} \cdot \overline{L_{f(t_1) \rightarrow x_0}} \dots L_{f(t_n) \rightarrow x_n} \cdot \overline{L_{f(t_{n+1}) \rightarrow x_n}} \\ &\simeq L_{f(0) \rightarrow x_0} \cdot \overline{L_{f(t_1) \rightarrow x_0}} \dots L_{f(t_n) \rightarrow x_n} \cdot \overline{L_{f(1) \rightarrow x_n}} \end{aligned}$$

■

Problem 4b. Show this applies in particular when X is open or when X is a union of finitely many closed convex sets

Proof. Let $B_\epsilon(x)$ be the epsilon ball around x for all $x \in X$ and $\forall \epsilon > 0$.

If X is open, since $X \subseteq \mathbb{R}^n$, $\forall x \in X$, $\exists \epsilon > 0 \ni x \in B_\epsilon(x) \subseteq X$. $B_\epsilon(x)$ is convex, thus between any two points in $B_\epsilon(x)$ the line connecting them would be in $B_\epsilon(x)$. Thus $\forall y \in B_\epsilon(x)$, the line between y and x is in $B_\epsilon(x)$. Thus $B_\epsilon(x)$ is star-shaped. Thus X is star-shaped.

If $X = \bigcup_{i=1}^m A_i$ where A_i is a closed convex set. $\forall x \in X$, $x \in A_i$ for some $i \in [1, \dots, m]$. WLOG, let $A_1, \dots, A_m$ be the subcollection of A_i that contain x . Thus $x \notin \bigcup_{i=m+1}^m A_i$. By closedness, $\exists d > 0 \ni \inf \{d(x, y) \mid y \in \bigcup_{i=m+1}^m A_i\} > d > 0$. Thus $B(x, \frac{d}{2}) \cap X \subseteq \bigcup_{i=1}^m A_i$.

$\forall y \in B(x, \frac{d}{2}) \cap X$. $B(x, \frac{d}{2})$ is convex so the line from x to y is in $B(x, \frac{d}{2})$. Furthermore $y \in \bigcup_{i=1}^m A_i \implies \exists j \ni m \geq j \geq 1$ such that $y \in A_j$. Since A_j is convex and $x \in A_j$, by definition of convexity the line from x to y is in A_j . Thus the line from x to y is in $B(x, \frac{d}{2}) \cap X$. Thus $B(x, \frac{d}{2}) \cap X$ is a star-shaped neighborhood. □

Problem 5.

Solution. 1. (a) $\implies$ (b): $\forall f : S^1 \rightarrow X$ continuous maps. By assumption, $f \simeq c$, thus $\exists F : S^1 \times I \rightarrow X$ a homotopy where $F(x, t) = f_t(x)$ and $f_0(x) = f$ and $f_1(x) = c(x) = x_0$. For D^2 use polar coordinates ($\theta \in [0, 2\pi]/(2\pi \sim 1)$ and $r \in [0, 1]$). Let $g : D^2 \rightarrow X$ where $g(r, \theta) = F(\frac{\theta}{2\pi}, 1 - r)$, g is continuous. $g(1, \theta) = F(\frac{\theta}{2\pi}, 0) = f(\frac{\theta}{2\pi}) \implies g|_{S^1} = f$.

2. (b) $\implies$ (c): Suppose (b) is true. There is a quotient map, $\pi : I \rightarrow S^1$ which glues the endpoints together taking them to $y_0 \in S^1$. $\forall x_0 \in X$ and $\forall f \in \pi_1(X, x_0)$, f factors through π where $f = g \circ \pi$ for some $g : S^1 \rightarrow X$ where $g(y_0) = x_0$. By (b), $\exists \tilde{g} : D^2 \rightarrow X$ which extends

g . There is a deformation retraction of D^2 onto y_0 , $F : D^2 \times I \rightarrow D^2$. Let $i : S^1 \hookrightarrow D^2$ be the natural inclusion. Let $G : I^2 \rightarrow X$ where $G(s, t) = g(F(i(\pi(s)), t))$. Thus $\forall s \in I$, $G(s, 0) = g(F(i(\pi(s)), 0)) = g(i(\pi(s))) = f(s)$ and $G(s, 1) = g(F(i(\pi(s)), 1)) = g(y_0) = x_0$. Furthermore, $\forall t \in I$ $G(0, t) = g(F(i(\pi(0)), t)) = g(y_0) = x_0$ and $G(1, t) = g(F(i(\pi(1)), t)) = x_0$. Thus $f \simeq c_{x_0}$. Thus $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

3. (c) $\implies$ (a): Suppose (c) is true. There is a quotient map, $\pi : I \rightarrow S^1$ which glues the endpoints together taking them to $y_0 \in S^1$. $\forall g : S^1 \rightarrow X$, $g \circ \pi \in \pi_1(X, g(y_0))$. Since $\pi_1(X, g(y_0)) = 0 \implies g \circ \pi \simeq c_{g(y_0)}$. Thus there is a homotopy $H : I^2 \rightarrow X$ which takes $g \circ \pi \simeq c_{g(y_0)}$ which fixes the endpoints. H factors through $F : S^1 \times I \rightarrow X$ where $H = F \circ (\pi \times I)$ where F is a homotopy between g and the constant map.

Suppose X is simply connected. Then $\pi_1(X, x_0) = 0$ for all $x_0 \in X$. Thus by (c) $\implies$ (a), all maps $S^1 \rightarrow X$ are homotopic to the constant map.

Suppose all maps $S^1 \rightarrow X$ are homotopic to the constant map. Then by $a \implies b$ and $b \implies c$, $\forall x_0 \in X$, $\pi_1(X, x_0) = 0$. $\forall x \in X$, let $c_x : S^1 \rightarrow X$ be the constant map. $\forall x, y \in X$ $c_x \simeq c_y$, thus exists $H : S^1 \times I \rightarrow X$. Take $f : I \rightarrow X$ where $f(t) = H(s_0, t)$. $f(0) = x$ and $f(1) = y$. Thus there is a path from x to y . Thus X is path connected. Thus X is simply connected. $\blacksquare$

Problem 6. We can regard $\pi_1(X, x_0)$ as the set of basepoint-preserving homotopy classes of maps $(S^1, s_0) \rightarrow (X, x_0)$. Let $[S^1, X]$ be the set of homotopy classes of maps $S^1 \rightarrow X$, with no conditions on basepoints. Thus there is a natural map $\varphi : \pi_1(X, x_0) \rightarrow [S^1, X]$ obtained by ignoring basepoints. Show that φ is onto if X is path-connected, and that $\varphi([f]) = \varphi([g])$ iff $[f]$ and $[g]$ are conjugate in $\pi_1(X, x_0)$. Hence φ induces a one-to-one correspondence between $[S^1, X]$ and the set of conjugacy classes in $\pi_1(X, x_0)$, when X is path-connected.

Solution. Suppose X is path-connected. $\forall g \in [S^1, X]$, let $x_1 = g(s_0)$ and let $h : [0, 1] \rightarrow X$ be the path that takes x_0 to x_1 . Let $f : I \rightarrow X$ be the loop with the base point x_1 which factors through g such that $f = g \circ \pi$. Let $h : [0, 1] \rightarrow X$ be the path from x_0 to x_1 . Let $h_t : [0, 1] \rightarrow X$ where $s \rightarrow h((1-s)t + s)$. Let $F : I^2 \rightarrow X$ where $(s, t) \rightarrow (h_t \cdot f \cdot \bar{h}_t)(s)$. Thus

$$F(s, t) := (h_t \cdot f \cdot \bar{h}_t)(s) = \begin{cases} h((1-2s)t + 2s) & \text{if } s \in [0, \frac{1}{2}] \\ f(4s-2) & \text{if } s \in [\frac{1}{2}, \frac{3}{4}] \\ h((4s-3)t + (1-(4s-3))) & \text{if } s \in [\frac{3}{4}, 1] \end{cases}$$

$F(0, t) = h(t)$, $F(\frac{1}{2}, t) = x_1$, $F(\frac{3}{4}, t) = x_1$, and $F(1, t) = h(t)$. Thus $F(\cdot, t)$ is a loop at $h(t)$.

$$F(s, 0) = \begin{cases} h(2s) & \text{if } s \in [0, \frac{1}{2}] \\ f(4s-2) & \text{if } s \in [\frac{1}{2}, \frac{3}{4}] \\ h(4-4s) & \text{if } s \in [\frac{3}{4}, 1] \end{cases}$$

Thus $F(s, 0) = (h \cdot f \cdot \bar{h})(s)$ and $h \cdot f \cdot \bar{h} \in \pi_1(X, x_0)$ and

$$F(s, 1) = \begin{cases} h(1) & \text{if } s \in [0, \frac{1}{2}] \\ f(4s-2) & \text{if } s \in [\frac{1}{2}, \frac{3}{4}] \\ h(1) & \text{if } s \in [\frac{3}{4}, 1] \end{cases}$$

Thus $F(s, 1) = (c_{x_1} \cdot f \cdot c_{x_1})(s)$. Thus,

$$f \simeq c_{x_1} \cdot f \cdot c_{x_1} \simeq h \cdot f \cdot \bar{h}$$

Thus $\varphi(h \cdot f \cdot \bar{h}) = g$. Thus φ is surjective.

Suppose $\varphi([f]) = \varphi([g])$, then there is a homotopy $F : I^2 \rightarrow X$ that gives $f \simeq g$ that doesn't preserve the basepoints, let $F_t(s) = F(s, t)$ where $F_0 = f$ and $F_1 = g$. Let $h : I \rightarrow X$ be $h(t) = F(0, t)$ be the map that shows how the base point gets transformed over the homotopy F . Since f and g have the same base point, x_0 , h is a loop about x_0 thus $h \in \pi_1(X, x_0)$. Furthermore, note that since $\forall t$, $F(\cdot, t)$ must be a loop, so $h(t) = F(1, t)$. Let $h_t : I \rightarrow X$ where $h_t(s) = h(st)$. Let $G : I^2 \rightarrow X$ where $G(s, t) = (h_t \cdot F_t \cdot \bar{h}_t)$. Note, $G(s, 0) = (c_{x_0} \cdot f \cdot \bar{c}_{x_0})(s)$ because $h_0(s) = h(0s) = h(0) = x_0$. Furthermore, $G(s, 1) = (h \cdot g \cdot \bar{h})(s)$ because $h_1(s) = h(1s) = h(s)$. Finally $G(0, t) = (h_t \cdot F_t \cdot \bar{h}_t)(0) = h_t(0) = x_0$ and $G(1, t) = (h_t \cdot F_t \cdot \bar{h}_t)(1) = h_t(1) = x_0$. Also note $F(0, t) = h(t) = G(\frac{1}{2}, t) = F(0, t)$ and $F(1, t) = G(\frac{3}{4}, t) = \bar{h}_t(0) = h_t(1) = h(t) = F(0, t) = F(1, t)$. Thus G is a valid homotopy that preserves base points of $\bar{c}_0 \cdot f \cdot \bar{c}_0 \simeq h \cdot g \cdot h$. Since $f \simeq \bar{c}_0 \cdot f \cdot \bar{c}_0$, $f \simeq h \cdot g \cdot h$.

Let $G : I^2 \rightarrow X$ where $G(s, t)$

Suppose $f : S^1 \rightarrow X$, let $[f]_*$ be the equivalence class of f in $[S^1, X]$ so homotopic without regard for basepoint in the homotopy. Let $f_i : S^1 \rightarrow X$, suppose f_2 starts at angle ρ in sequence $f_1 \dots f_n$. Then $G : S^1 \times I \rightarrow X$ where $G(\theta, s) = (f_1 \dots f_n)(\theta + s\rho)$. $G(\theta, 0) = (f_1 \dots f_n)(\theta)$ and $G(\theta, 1) = (f_1 \dots f_n)(\theta + \rho) = ((f_2 \dots f_n) \cdot f_1)(\theta) \simeq (f_2 \dots f_n \cdot f_1)(\theta)$. Thus $[f_1 \dots f_n]_* = [f_2 \dots f_n \cdot f_1]_*$. This argument can be extended to all cyclic permutations of $f_1 \dots f_n$.

Suppose $[f]$ and $[g]$ be conjugate. Thus $f \simeq h \cdot g \cdot \bar{h}$ for some $[h] \in \pi_1(X, 1)$. Thus $[f]_* = \varphi([f]) = \varphi([h \cdot g \cdot \bar{h}]) = [h \cdot g \cdot \bar{h}]_*$, by the argument above. $\varphi([f]) = [h \cdot g \cdot \bar{h}]_* = [g \cdot h \cdot \bar{h}]_* = [g]_* = \varphi([g])$. Thus $\varphi([f]) = \varphi([g])$. ■

Problem 7.

Solution. $F : S^1 \times I \times I \rightarrow S^1 \times I$ where

$$F(\theta, s, t) := (\theta - 2\pi s(1 - t), s)$$

Thus $\forall(\theta, s) \in S^1 \times I$, $F(\theta, s, 0) = (\theta - 2\pi s, s) = f(\theta, s)$ and $F(\theta, s, 1) = (\theta - 2\pi(0), s) = (\theta, s) = 1_{S^1 \times I}(\theta, s)$. Thus F is the homotopy that gives you $f \simeq 1_{S^1 \times I}$. Furthermore, $\forall(\theta, t) \in S^1 \times I$, $F(\theta, 1, t) = (\theta - 2\pi(1 - t), 1)$ and $F(\theta, 0, t) = (\theta - 2\pi(0)(1 - t), 0) = (\theta, s) = 1_{S^1 \times I}(\theta, s)$. Thus F keeps the $S^1 \times \{0\}$ stationary throughout the homotopy and varies $S^1 \times \{1\}$ throughout the homotopy. ■

Problem 8. Does the Borsuk-Ulam theorem hold for the torus? In other words, for every map $f : S^1 \times S^1 \rightarrow \mathbb{R}^2$ must there exist $(x, y) \in S^1 \times S^1$ such that $f(x, y) = f(-x, -y)$?

Proof. Let $i : S^1 \times S^1 \hookrightarrow \mathbb{R}^3$, be the natural inclusion of the torus into $\mathbb{R}^3$ where the center of the torus is the origin. Let $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the the projection of $\mathbb{R}^3$ onto $\mathbb{R}^2$. Let $f : S^1 \times S^1 \rightarrow \mathbb{R}^2$ where $f = \pi \circ i$. Then $\forall p \in S^1 \times S^1$, $f(-p) = -f(p)$ and since $f(p) \neq (0, 0)$ then $f(-p) = -f(p) \neq f(p)$ for all $p \in S^1 \times S^1$. Thus Borsuk - Ulam doesn't hold for the Torus. □

Lemma C. For $s \in \mathbb{R}$ and $v \in S^3$, let $P_{v,s}$ is a affine plane that pass through sv with a normal vector v . Let A be a compact subset of $\mathbb{R}^3$. For $v \in S^2$, $\exists P_v$ a plane in $\mathbb{R}^3$ that divides A into two pieces of equal measure.

Proof. For any affine plane P_{v,x_0} , let $P_{v,x}^+ = \bigcup_{t \geq 0} P + vt$ so everything "above" P .

For any affine plane P_{v,x_0} , let $P_{v,x}^- = \bigcup_{t \geq 0} P - vt$ so everything "below" P . Note $P_{v,x}^- \cup P_{v,x}^+ = \mathbb{R}^3$.

$\forall v \in S^2$, let L_v be the line determined by v . A is bounded. P_v^+ is a closed subspace of $\mathbb{R}^3$. Since A is bounded, there is a $t_0, t_1 \in \mathbb{R}$ such that $\text{vol}(P_{v,t_0}^+ \cap A) = 0$ and $\text{vol}(P_{v,t_1}^+ \cap A) = \text{vol}(A)$. We can continuously shift the plane along the line and the volume should change continuously. By the intermediate value theorem, $\exists s \in \mathbb{R} \ni t_1 \leq s \leq t_0 \ni \text{vol}(P_{v,s}^+ \cap A) = \frac{1}{2} \text{vol}(A)$ which would mean $\text{vol}(P_{v,s}^- \cap A) = \frac{1}{2} \text{vol}(A)$. Thus $P_{v,s}$ splits A in 2 pieces of equal measure. $\square$

Problem 9. Let A_1, A_2, A_3 be compact sets in $\mathbb{R}^3$. Use the Borsuk-Ulam theorem to show that there is one plane $P \subseteq \mathbb{R}^3$ that simultaneously divides each A_i into two pieces of equal measure.

Solution. $\forall v \in S^2$, by Lemma B, $\exists P_{v,s_{i,v}}$ that splits A_i into peices of equal measures. $s_{i,v} - s_{j,v}$ is the distance between the planes $P_{v,s_{i,v}}$ and $P_{v,s_{j,v}}$. Additionally note that $s_{i,-v} = -s_{i,v}$.

Let $f : S^2 \rightarrow \mathbb{R}^2$ take $v \rightarrow (s_{2,v} - s_{1,v}, s_{3,v} - s_{1,v})$. This is a continous map because a small deviation in a unit vector v in S^2 will produce small deviations in the planes with normal vectors v that divide A_i and hence small deviations in the distances between affine planes that divide A_i , and hence small deviations in f . Thus f is continous. By the Borsuk Ulam Theorem, theorem $\exists v \in S^3$ such that

$$(s_{2,v} - s_{1,v}, s_{3,v} - s_{1,v}) = f(v) = f(-v) = (s_{1,v} - s_{2,v}, s_{1,v} - s_{3,v})$$

Thus $s_{2,v} - s_{1,v} = s_{1,v} - s_{2,v} \implies s_{1,v} - s_{2,v} = 0$ and $s_{1,v} - s_{3,v} = s_{3,v} - s_{1,v} \implies s_{3,v} - s_{1,v} = 0$. Thus the distances between the planes are 0. Thus $P_{v,s_{v,1}}, P_{v,s_{v,2}}, P_{v,s_{v,3}}$ are the same plane P . Thus P divides each A_i into peices of equal measure. $\blacksquare$

Problem 10.

Solution. $\forall f \in \pi_1(X \times \{y_0\}, (x_0, y_0))$ and $\forall g \in \pi_1(\{x_0\} \times Y, (x_0, y_0))$. Let $p_X : X \times Y \rightarrow X$ and $p_Y : X \times Y \rightarrow Y$ be the standard projection maps. Let $F : I^2 \rightarrow X \times Y$ where

$$F(s, t) := \begin{cases} g(2s) & \text{if } s \in [0, \frac{1}{2}t] \\ (p_X(f(2s - t), p_Y(g(t))) & \text{if } s \in [\frac{1}{2}t, \frac{1}{2}t + \frac{1}{2}] \\ g(2s - 1) & \text{if } s \in [\frac{1}{2}t + \frac{1}{2}, 1] \end{cases}$$

Let's check that F is well defined. When $s = \frac{1}{2}t$,

$$g(t) = (x_0, p_Y(g(t))) = \left(p_X \left(f \left(2 \left(\frac{1}{2}t \right) - t \right), p_Y(g(t)) \right) \right) = (p_X(f(0)), p_Y(g(t))) = (x_0, p_Y(g(t))).$$

When $s = \frac{1}{2}t + \frac{1}{2}$,

$$\begin{aligned} (p_X(f(2(0.5t + 0.5) - t), p_Y(g(t)))) &= (p_X(f(1), p_Y(g(t)))) \\ &= (x_0, p_Y(g(t))) \\ &= g(t) \\ &= g(2(0.5t + 0.5) - 1) \end{aligned}$$

Let's make sure F preserves endpoints,

$$F(0, t) = g(0) = (x_0, y_0)$$

and

$$F(1, t) = g(1) = (x_0, y_0)$$

$\forall s \in I,$

$$\begin{aligned} F(s, 0) &= \begin{cases} g(2s) & \text{if } s \in [0, 0] \\ (p_X(f(2s-0), p_Y(g(0)))) & \text{if } s \in [0, \frac{1}{2}] \\ g(2s-1) & \text{if } s \in [\frac{1}{2}, 1] \end{cases} \\ &= \begin{cases} (x_0, y_0) & \text{if } s \in [0, 0] \\ (p_X(f(2s), y_0)) & \text{if } s \in [0, \frac{1}{2}] \\ g(2s-1) & \text{if } s \in [\frac{1}{2}, 1] \end{cases} \\ &= \begin{cases} f(2s) & \text{if } s \in [0, \frac{1}{2}] \\ g(2s-1) & \text{if } s \in [\frac{1}{2}, 1] \end{cases} \end{aligned}$$

and

$$\begin{aligned} F(s, 1) &= \begin{cases} g(2s) & \text{if } s \in [0, \frac{1}{2}] \\ (p_X(f(2s-1), p_Y(g(1)))) & \text{if } s \in [\frac{1}{2}, 1] \\ g(2s-1) & \text{if } s \in [1, 1] \end{cases} \\ &= \begin{cases} g(2s) & \text{if } s \in [0, \frac{1}{2}] \\ (p_X(f(2s-1), y_0)) & \text{if } s \in [\frac{1}{2}, 1] \\ (x_0, y_0) & \text{if } s \in [1, 1] \end{cases} \\ &= \begin{cases} g(2s) & \text{if } s \in [0, \frac{1}{2}] \\ f(2s-1) & \text{if } s \in [\frac{1}{2}, 1] \end{cases} \end{aligned}$$

Thus F is the homotopy that gives $f \cdot g = g \cdot f$. Thus $[f][g] = [fg] = [gf] = [g][f]$. ■